complement

Something that completes or fills up a whole.

See complementary addition, complementary angles

complementary addition

(i) Finding the amount to complete a set.

Example

What has to be added to seven to make ten?

$7 + \square = 10$

$7 + 3 = 10$

Answer: Three has to be added.

(ii) Counting on to a higher total (as change is given after a purchase).

Example

Shopping costs $17.50. I pay with a $20 note. I get $2.50 change. This is evaluated by finding what must be added to $17.50 to make $20.

(iii) The method of 'subtracting' which converts the subtraction question to an addition question.

Example

$$21 - 19 = 2$$

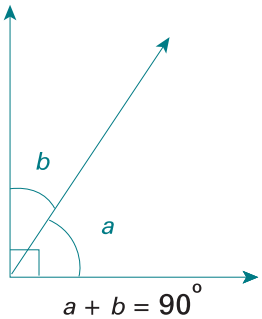
Instead of taking nineteen away from twenty-one we think how much must be added to nineteen to make twenty-one.

See addition, set, subtraction

complementary angles

Two angles that together measure 90°.

Example



$\angle a$ and $\angle b$ are complementary.

$\angle a$ is the complement of $\angle b$.

$\angle b$ is the complement of $\angle a$.

See supplementary angles

complex fraction

A fraction whose numerator, denominator, or both, are fractions.

Examples

$$\frac{\frac{1}{2}}{5} \qquad \frac{3}{\frac{4}{7}} \qquad \frac{\frac{1}{2}}{\frac{3}{4}} \qquad \frac{\frac{a}{b}}{\frac{c}{d}}$$

Note: To simplify a complex fraction means the same as division of fractions. It can be done in two ways.

Example

Divide $\frac{1}{2}$ by $\frac{2}{3}$.

1 $\frac{1}{2} \div \frac{2}{3} = \frac{1}{2} \times \frac{3}{2} = \frac{3}{4}$

2 $\frac{\frac{1}{2}}{\frac{2}{3}} = \frac{1 \times 3}{2 \times 2} = \frac{3}{4}$

composite number

A number with factors other than itself and one.

Examples

$12 = 12 \times 1 = 3 \times 4 = 3 \times 2 \times 2$

$33 = 33 \times 1 = 3 \times 11$

Both twelve and thirty-three are composite numbers.

$17 = 17 \times 1 \quad 23 = 23 \times 1$

Seventeen and twenty-three are not composite numbers.

Numbers which have no other factors except themselves and one, such as seventeen, are called prime numbers.

Every whole number greater than one is either:

(i) a prime number
(2, 3, 5, 7, 11 …)

or

(ii) a composite number
(4, 6, 8, 9, 10, 12, 14 …).

See factors, prime number